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Data Driven Learning and Control Project Discussions



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Project 1: Waymo Prediction Challenge

Challenge Description:

Model multi-modal future behaviors under uncertainty

Task:

Predict future trajectories of traffic participants

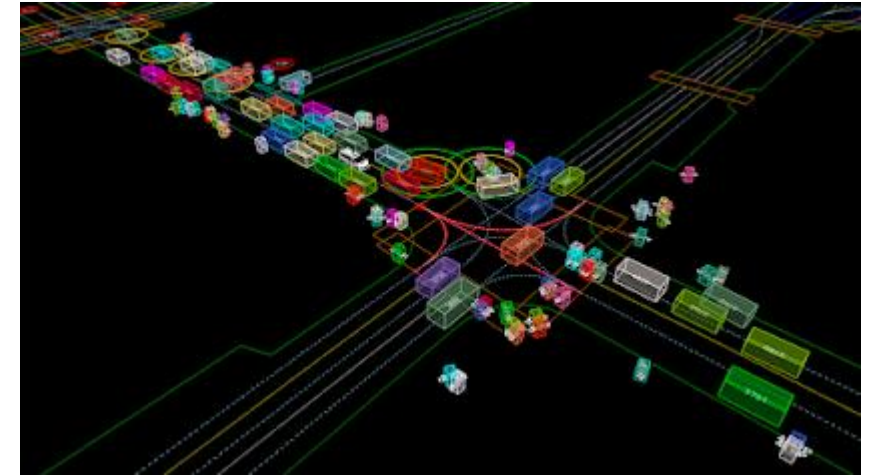
- Ego agents are vehicles, pedestrians, cyclists
- Prediction window: **8-second horizon** (80 steps at 10 Hz)
- using **1 second of past motion history** and high-definition map context.

Evaluation metrics:

<https://waymo.com/open/challenges/2024/motion-prediction/>

Dataset:

<https://waymo.com/open/data/motion/>



Project 2: Comma AI Controls Challenge

Challenge Description:

Design and evaluate learning-based vehicle controllers for lateral and longitudinal control using real driving data.

Task:

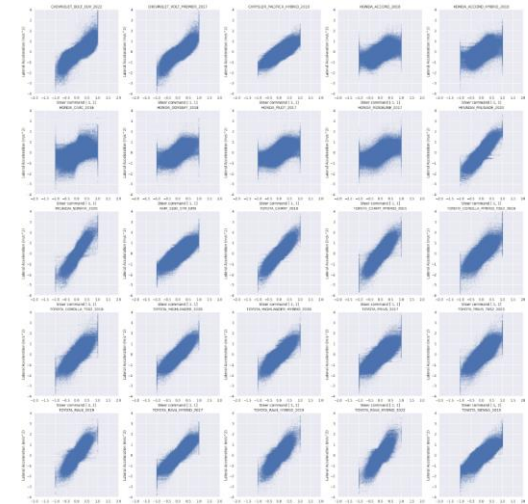
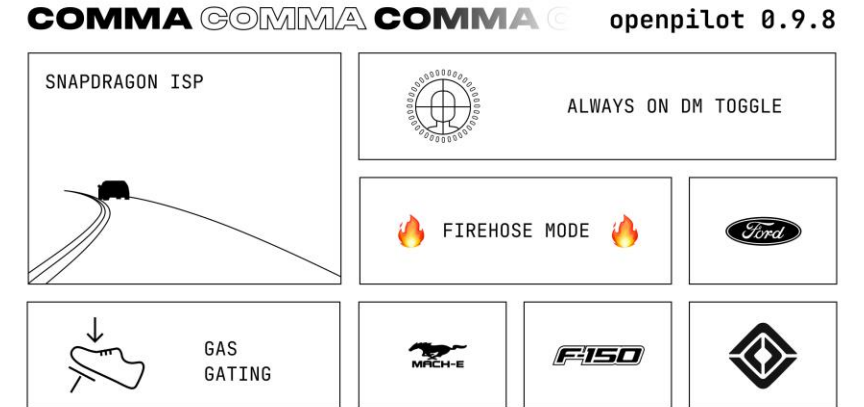
- Use the provided dataset/simulator to identify a data-driven vehicle dynamics model
- Design controllers (MPC, RL, PID, etc.)
- Generate control commands (steering, throttle, brake) to track a reference path.

Evaluation Metrics:

https://github.com/commaai/controls_challenge

Dataset:

https://github.com/commaai/controls_challenge



Project 3: Offroad Driving (Modeling)

Project Description:

Learn predictive models for vehicle motion in unstructured off-road environments with complex terrain and sensing noise.

Task:

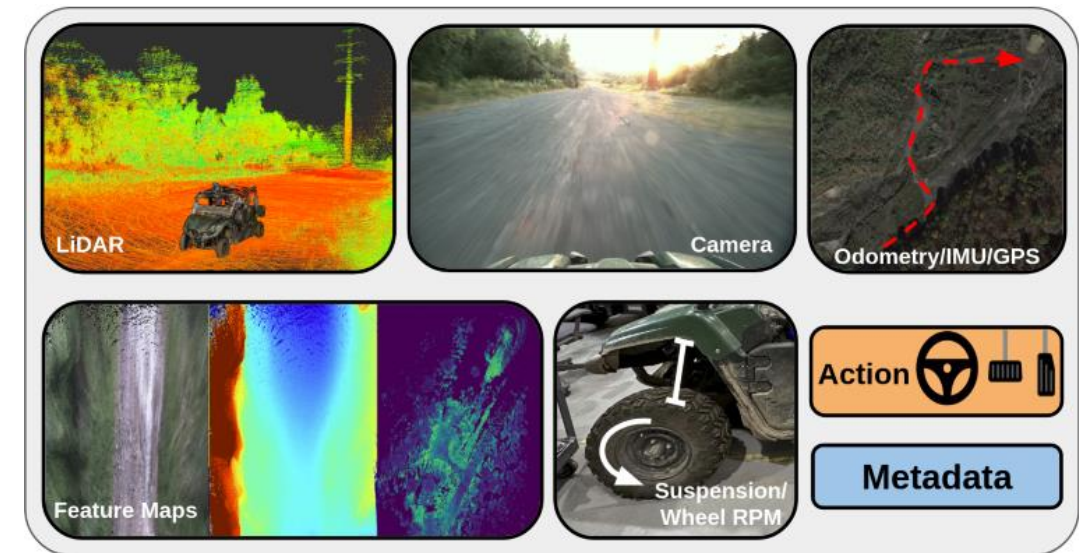
- Predict future vehicle state (pose, velocity) over a fixed horizon
- Use **multi-modal inputs**:
 - Camera
 - LiDAR
 - IMU / proprioception
- Handle **terrain-induced dynamics** (slip, bumps, elevation changes)

Evaluation Metrics:

Design your own based on prediction accuracy

Dataset:

<https://theairlab.org/TartanDrive2/>



Project 4: Offroad Driving 2 (Control Design)

Project Description:

Design controllers using data-driven models for vehicle motion in unstructured off-road environments.

Task:

- Design terrain informed data-driven models (using cameras, LiDAR and IMU data)
- Design your favorite controller (MPC, RL)
- Handle **terrain-induced dynamics** (slip, bumps, elevation changes)

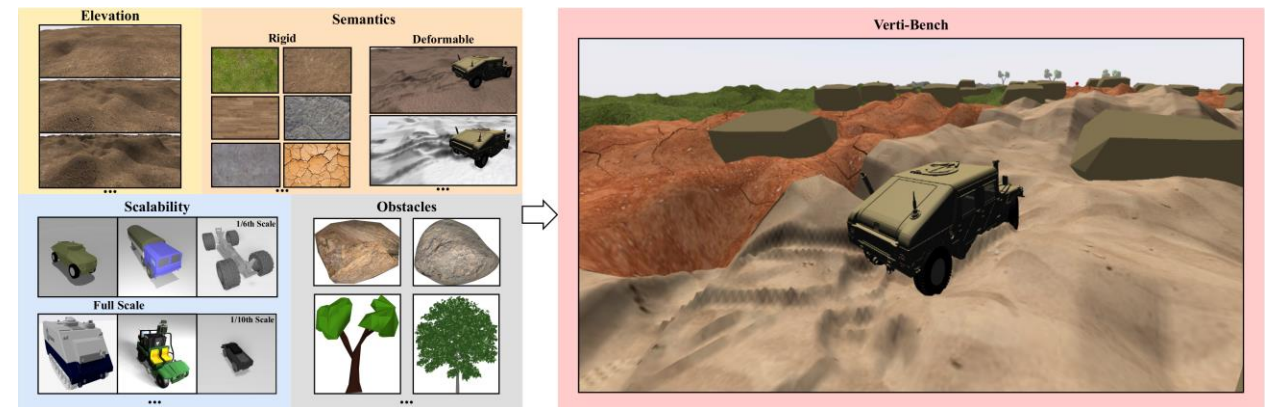


Evaluation Metrics:

Design your own based on control performance

Dataset and Simulator:

https://github.com/RobotiXX/verti_bench



Project 5: Agile Quadrotors (open ended)

Project Description:

Learn a data-driven dynamical model of an agile quadrotor and use it to **design a state estimator** for high-speed flight.

Task:

- Use vision and inertial data to learn a data-driven model
- Design your favorite state estimator using the learned model:
 - EKF / UKF with learned dynamics
 - Optimization-based filtering or smoothing
 - Learning-augmented estimators
- Estimate full state (pose, velocity, orientation) during **aggressive maneuvers**

Evaluation Metrics:

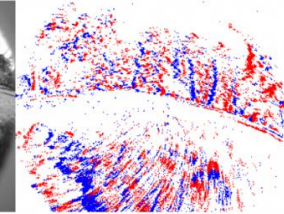
Design your own based on estimation accuracy

Dataset:

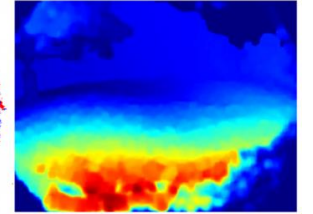
<https://fpv.ifi.uzh.ch/>



(a) Outdoor sequence 15



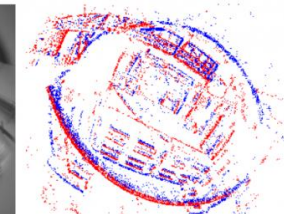
(b) Events



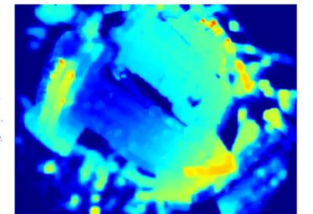
(c) Optical flow



(d) Indoor sequence 6



(e) Events



(f) Optical flow



Project 6: Quadruped Locomotion (open ended)

Project Description:

Learn predictive models for quadruped locomotion using real-world sensory and proprioceptive data.

Task:

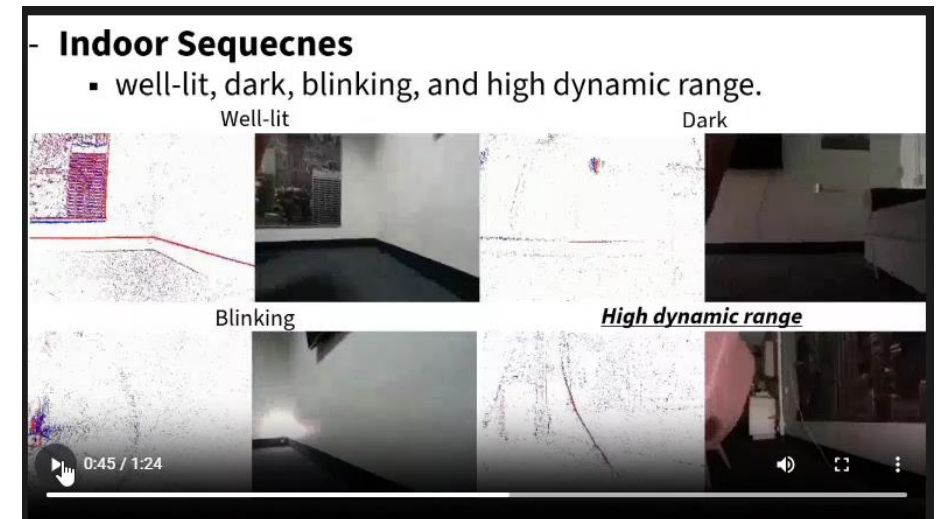
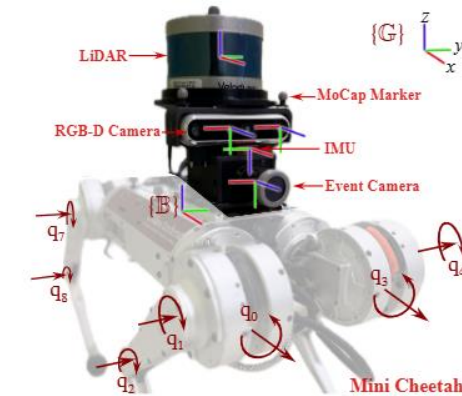
- Predict future robot state (base pose, velocity, joint states) over a fixed horizon
- Use **robot-centric sensing**:
 - Joint positions, velocities, torques
 - IMU (orientation, angular velocity)
 - Terrain/contact information
- Capture **contact-rich, hybrid dynamics** including backflips

Evaluation Metrics:

Design your own based on prediction accuracy

Dataset:

<https://daroslab.github.io/cear/>



Questions?

